

LO4 -- Kubernetes Practical Revision Guide

(Expanded)

This guide follows the LO4 topics from the uploaded learning outcomes and explains **what to do, why you do it, and what you should expect to see**.

1. Deployments

A Deployment manages stateless applications. It automatically creates ReplicaSets, maintains the desired number of Pods, performs rolling updates and self-healing.

Example:

```
kubectl create deployment web --image=nginx:1.25
```

```
kubectl get deployments
```

```
kubectl get rs
```

```
kubectl get pods
```

Export YAML:

```
kubectl get deployment web -o yaml > web.yaml
```

2. Scaling

Imperative:

```
kubectl scale deployment web --replicas=5
```

Declarative: Edit **replicas** in YAML then:

```
kubectl apply -f web.yaml
```

Verify:

```
kubectl get deploy
```

```
kubectl get rs
```

3. Rolling Updates

```
kubectl set image deployment/web nginx=nginx:1.27
```

kubectl rollout status deployment/web

Benefits: - Minimal downtime - Gradual replacement - Easy rollback

Rollback:

kubectl rollout undo deployment/web

kubectl rollout history deployment/web

4. Deployment Strategies

RollingUpdate

- Default
- High availability
- Uses **maxSurge** and **maxUnavailable**

Recreate

- Deletes old Pods before creating new ones.
- Useful when only one instance may access a database or shared files.

5. Workload Types

Deployment - Stateless applications - Web servers - APIs

StatefulSet - Databases - Stable pod names - Persistent storage

DaemonSet - One Pod per node - Monitoring agents - Log collectors

Job - One-time batch work

CronJob - Scheduled execution

6. Storage

StorageClass → defines storage type.

PersistentVolume (PV) - Actual storage resource.

PersistentVolumeClaim (PVC) - Request for storage.

emptyDir - Temporary storage deleted with Pod.

ConfigMap - Non-sensitive configuration.

Secret - Sensitive information (passwords, certificates).

7. Services

ClusterIP - Internal communication.

NodePort - External access via nodeIP:port.

LoadBalancer - Cloud load balancer.

Headless Service - No virtual IP. - Used with StatefulSets.

8. Networking & DNS

Every Pod gets its own IP.

Services provide stable DNS:

service.namespace.svc.cluster.local

9. Health Probes

Liveness: Restarts unhealthy container.

Readiness: Prevents traffic until application is ready.

Startup: Allows slow applications extra startup time.

10. Common Commands

kubectl get all

kubectl describe pod POD

kubectl logs POD

kubectl logs POD --previous

kubectl exec -it POD -- sh

kubectl get events

kubectl get endpoints

kubectl rollout status

kubectl rollout undo

kubect! port-forward

Exam Checklist

Know the differences between: - Deployment vs StatefulSet - Job vs CronJob - ConfigMap vs Secret - ClusterIP vs NodePort vs LoadBalancer - Liveness vs Readiness vs Startup - PV vs PVC vs StorageClass

Always verify: - Pods Running - READY column - Events - Logs - Endpoints - Labels/selectors